

Causal Inference HW1

1. A public health researcher wants to estimate the causal effect of regular coffee consumption (C) on the risk of heart disease (H). The researcher suspects that smoking (S) is associated with both coffee drinking and heart disease. They also believe that an individual's stress level (T) can lead them to both smoke more and drink more coffee.
 - a. Draw a DAG that represents the causal relationships described among the 4 variables.
 - b. Based on your DAG, is S a confounder, a mediator, or neither? Explain your reasoning.
 - c. To get an unbiased estimate of the causal effect of C on H , which variables must the researcher control for? Explain your answer.
2. A data scientist is running an A/B test to see if a new website button design ($T = 1$) increases user engagement (measured in clicks) compared to the old design ($T = 0$). Below is a table showing the potential outcomes for 10 users. $Y_i(1)$ is the number of clicks for user i if they see the new button, and $Y_i(0)$ is the number of clicks if they see the old button.

i	$Y_i(1)$	$Y_i(0)$	i	$Y_i(1)$	$Y_i(0)$
1	12	8	6	9	4
2	7	5	7	11	7
3	10	11	8	5	3
4	14	9	9	13	10
5	8	6	10	6	6

- a. What is the fundamental problem of causal inference in this context? Why can't we observe both potential outcomes for the same user at the same time?
- b. Calculate the individual treatment effect for each user.
- c. Calculate the Average Treatment Effect (ATE) for this entire population of 10 users. Interpret what this value means.
- d. Now, assume a real experiment was run where users 1-5 were shown the new button (treatment group) and users 6-10 were shown the old button (control group). Calculate the observed difference in means. How does this value compare to the true ATE you calculated? Explain any difference in terms of selection bias.